

Smart Companion

Version: 1.0

Status: Draft

Target Platform: Web Application (PWA Capable)

1. Executive Summary

The Smart Companion is an AI-powered life operating system designed to assist neurodiverse individuals (ADHD, Dyslexia, etc.) in overcoming "Task Paralysis" and "Time Blindness." Unlike standard to-do lists, this system adapts to the user's biological energy rhythm and visual processing needs, breaking overwhelming goals into "MicroWins."

2. Feature Set 1: The Calibration Wizard (Onboarding)

Goal: Capture the user's functional preferences and biological rhythm without using medical jargon or causing "form fatigue."

2.1 User Flow

Index Page -> Auth (Signup) -> **Calibration Wizard** -> Dashboard

2.2 Functional Requirements

Step 1: Visual Comfort Tuner (Dyslexia Support)

- **Objective:** Configure the UI to minimize visual stress.
- **Interaction:** A split-screen comparison.
 - *Panel A:* Standard Sans-Serif font, normal leading, white background.
 - *Panel B:* **OpenDyslexic** font, 1.5x line spacing, "Cream" background (Hex: #FFFDD0).
- **Question:** "Which side is calmer for your eyes?"
- **Output Variable:** visual_mode (Standard/Dyslexia-Friendly).

Step 2: Cognitive Load Calibration (Granularity)

- **Objective:** Determine how the AI should decompose tasks.
- **Interaction:** Two cards showing a task list.
 - *Card A (Low Granularity):* "1. Clean the Kitchen."
 - *Card B (High Granularity):* "1. Throw away trash on counter. 2. Put dirty dishes in sink. 3. Wipe the table."
- **Question:** "Which instruction feels less overwhelming?"

- **Output Variable:** granularity_preference (Macro/Micro).

Step 3: Chronotype Detection (Energy Scheduler)

- **Objective:** Map the user's peak energy hours.
- **Interaction:** A visual slider or three simple illustrated cards.
 - *Morning Lark:* "I'm sharpest before noon."
 - *Mid-Day:* "I peak in the afternoon."
 - *Night Owl:* "I get my best work done after dark."
- **Output Variable:** chronotype (Lark/Hummingbird/Owl).

Step 4: Sensory Profile (Gamification)

- **Objective:** Determine reward mechanisms.
- **Interaction:** Toggle switch.
 - *Gamified:* "I like streaks, badges, and sounds."
 - *Zen Mode:* "I prefer a quiet, calm interface."
- **Output Variable:** gamification_level (High/Low).

2.3 Technical Constraints

- **Privacy:** All answers are stored in the local SQLite NeuroProfile table.
- **Persistence:** Settings must be applied immediately to the Onboarding UI itself (e.g., if they pick Dyslexia font, the rest of the wizard switches to that font instantly).

3. Feature Set 2: The Dashboard (The Command Center)

Goal: Eliminate "Decision Fatigue" by hiding future tasks and focusing strictly on the "Now."

3.1 Design Layout (Wireframe Logic)

The screen is divided into three distinct zones to guide the eye.

Zone	Name	Purpose
Top Left	Energy Battery	Contextual awareness of the user's current state.
Center	The Focus Solo	The primary interaction point. Shows only ONE active step.

Bottom	The Brain Dump	Quick capture to offload working memory.
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3.2 Detailed Component Requirements

Component A: The Energy Battery

- **Logic:** Compares `Current_System_Time` vs. User's Chronotype.
- **Visual States:**
 - *High Charge (Green):* "Peak Energy. Great time for complex tasks."
 - *Low Charge (Orange):* "Energy Dip. Let's stick to easy wins."
- **Data Source:** Scheduler Algorithm (Python backend).

Component B: The Focus Solo (The "Task Paralysis" Killer)

- **Behavior:**
 - This component hides the full to-do list.
 - It fetches the **top priority MicroWin** from the backend.
 - *Example Display:* A large, clean card reading: **"Open your Physics Textbook."** (Not "Study Physics").
- **Actions:**
 - **"Done" Button:** Triggers a "Dopamine Hit" (visual ripple effect + subtle sound). Loads the next step.
 - **"Stuck" Button:** Summons the AI Companion to break the step down further (e.g., "Okay, just find the book first").
 - **"Defer" Button:** Pushes the task to the "Energy Dip" slot later in the day.

Component C: The Brain Dump

- **Behavior:** A constantly visible text input field at the bottom.
- **Function:** Users type raw thoughts here (e.g., "Need to call Mom").
- **Backend Logic:** These inputs are sent to the LLM later to be parsed, categorized, and scheduled. They do *not* clutter the immediate view.

Component D: Zero State (First Login)

- **Trigger:** User has no tasks.
- **UI:** A calm illustration (e.g., a resting landscape).
- **Copy:** "All caught up. What's on your mind for today?"
- **Call to Action:** Focus is placed on the "Brain Dump" input.

4. API Requirements (FastAPI Endpoints)

4.1 Onboarding

- POST /api/v1/profile/calibrate
 - **Input:** JSON object with preferences.
 - **Action:** Updates NeuroProfile in SQLite.

4.2 Dashboard

- GET /api/v1/dashboard/state
 - **Returns:**
 - energy_level: (High/Low)
 - current_microwin: { "text": "Open VS Code", "id": 101 }
 - streak_count: (Integer)
 - POST /api/v1/tasks/complete/{id}
 - **Action:** Marks task done, updates streak, returns next_microwin.
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5. Non-Functional Requirements

1. **Load Time:** Dashboard must load in < 1.5 seconds (critical for ADHD users who lose focus quickly).
2. **Accessibility:** Must pass **WCAG 2.1 AAA** standards (High contrast ratios, screen reader compatibility).
3. **Error Handling:** Never show red "Error" boxes (induces anxiety). Use gentle "Retry" prompts in neutral colors.

6. Success Metrics (For Judges)

- **Retention:** Do users return for Day 2? (Measured by "Streak" data).
- **Completion Rate:** % of "MicroWins" completed vs. "Big Goals" set.
- **Latency:** Time taken from "Brain Dump" input to "Task Scheduled" (Efficiency of AI).